Kenbunshoku

Project Documentation

Global AI Hackathon Series with Qwen Cloud — Track 5: EdgeAgent

1. Executive Summary

Kenbunshoku is a camera-agnostic security agent that upgrades any existing doorbell cam, webcam, or IP camera from a passive recorder into a context-aware watcher. It detects a person approaching, reasons about who they likely are using Qwen Cloud's vision-language model, checks that against a memory of prior visits, and sends the homeowner a plain-language alert. It never takes action on its own — the homeowner always decides.

2. Problem & Motivation

Traditional cameras answer "was there motion?" — not "should I care?" This produces two failure modes: alert fatigue (every passerby triggers a notification) and after-the-fact awareness (footage is only useful once reviewed, often after an incident). Kenbunshoku targets the gap between raw motion detection and genuine situational awareness.

3. Solution Overview

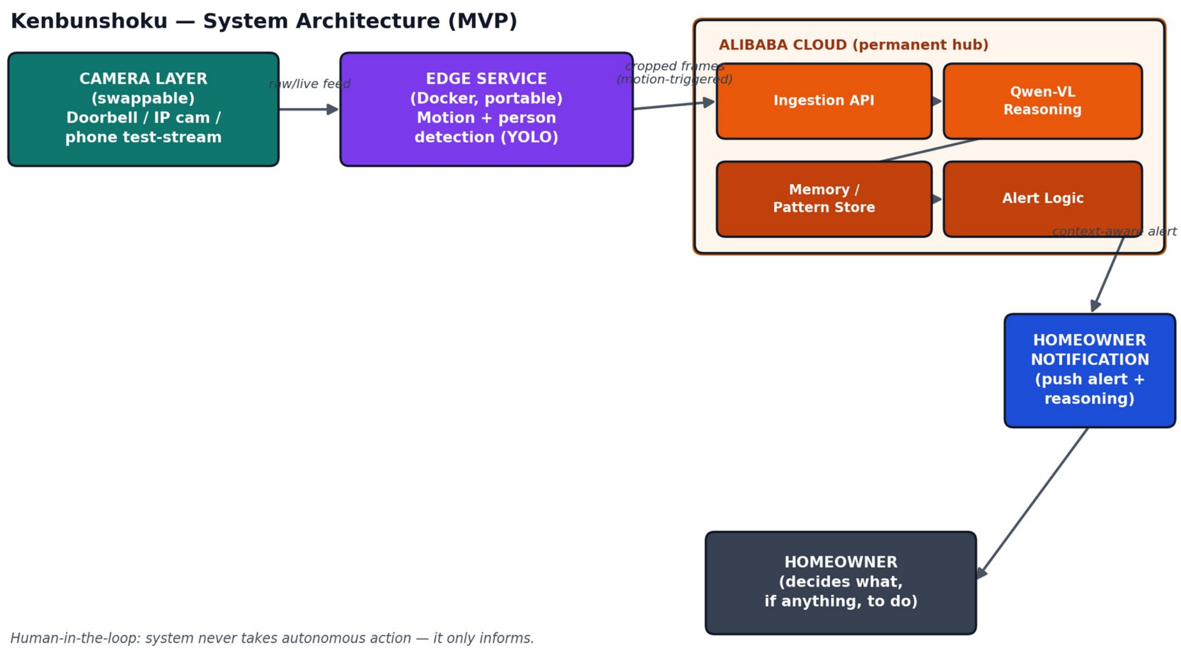
The system separates concerns into three swappable layers — camera, edge, and cloud — so it works with hardware people already own and scales to a real product beyond the hackathon prototype:

- Camera layer: any live video source (real doorbell/IP cam in production; a phone running an IP-camera app during MVP testing)
- Edge layer: a portable, containerized service that does cheap local person/motion detection before anything is sent to the cloud
- Cloud layer: Qwen-VL reasoning, memory of visit patterns, and alert logic, permanently hosted on Alibaba Cloud

4. Key Features

- Camera-agnostic ingestion — no proprietary hardware required
- Context classification, not just detection — distinguishes delivery-like, familiar, or anomalous visits
- Pattern memory — recognizes recurring visitors and routines over time, reducing noise the longer it runs
- Human-in-the-loop only — every output is an informative alert; the system never locks doors, sounds alarms, or contacts authorities on its own
- Bandwidth-aware — sends cropped, motion-triggered frames rather than continuous raw video
- Graceful degradation — falls back to a generic alert if cloud connectivity drops

5. System Architecture



The camera layer streams to the edge service, which performs lightweight local detection and forwards only relevant cropped frames to Alibaba Cloud. There, an ingestion API hands frames to Qwen-VL for reasoning, cross-referenced against the memory/pattern store, before alert logic pushes a notification to the homeowner — who remains the sole decision-maker.

6. Technology Stack

Layer	Tooling	Notes
Camera (test)	Phone (iPhone 12 / Galaxy S22) + IP-camera streaming app	Free; stand-in for a real doorbell/IP camera
Edge service	Docker, Python, YOLOv8n (open weights)	Free/open source; portable to any edge box
Cloud backend	Alibaba Cloud (ECS or Function Compute)	Required for hackathon deployment proof
Reasoning	Qwen-VL via Qwen Cloud API ($40 hackathon voucher / free trial)	OpenAI-compatible SDK
Memory store	Lightweight DB (SQLite/Redis) on Alibaba Cloud	Stores visit timestamps + context summaries
Notifications	Free push service (e.g., Firebase Cloud Messaging or ntfy.sh)	Client built in Flutter, matching existing skillset
Repo & docs	GitHub (public, open-source license)	Required for submission

7. Track Alignment

Primary: Track 5 — EdgeAgent. The system perceives via an edge sensor (camera + local detection), reasons via Qwen Cloud APIs, and acts locally by notifying — directly matching the track's ask for "robust edge-cloud orchestration under bandwidth/latency constraints, privacy-aware data handling, and graceful degradation." The memory/pattern layer additionally echoes Track 1 (MemoryAgent) as a supporting strength, without being pitched as a separate submission.

8. Ethical & Safety Considerations

- No autonomous action — the system only informs; the homeowner decides
- No weapon/threat classification in the MVP, to avoid unreliable high-stakes false positives
- Only motion-triggered, relevant crops are sent to the cloud — not continuous surveillance video
- Framed as "context and pattern awareness," not "crime prediction," to keep the tool's claims honest and its risk of false accusation low

9. Future Work (Post-MVP)

- Careful, properly-validated threat/weapon detection as an opt-in advanced feature
- Multi-camera fusion across a property
- Native integration with commercial doorbell/IP camera SDKs (Ring, Reolink, ONVIF-compliant devices)
- LiDAR/depth-based 3D approach-path prediction on supported devices

10. Open Source & Licensing

The project repository will be public with a permissive open-source license (e.g., MIT) visible in the repository's About section, per hackathon submission requirements. All tooling used is free or covered by hackathon-provided credits — no paid dependencies.